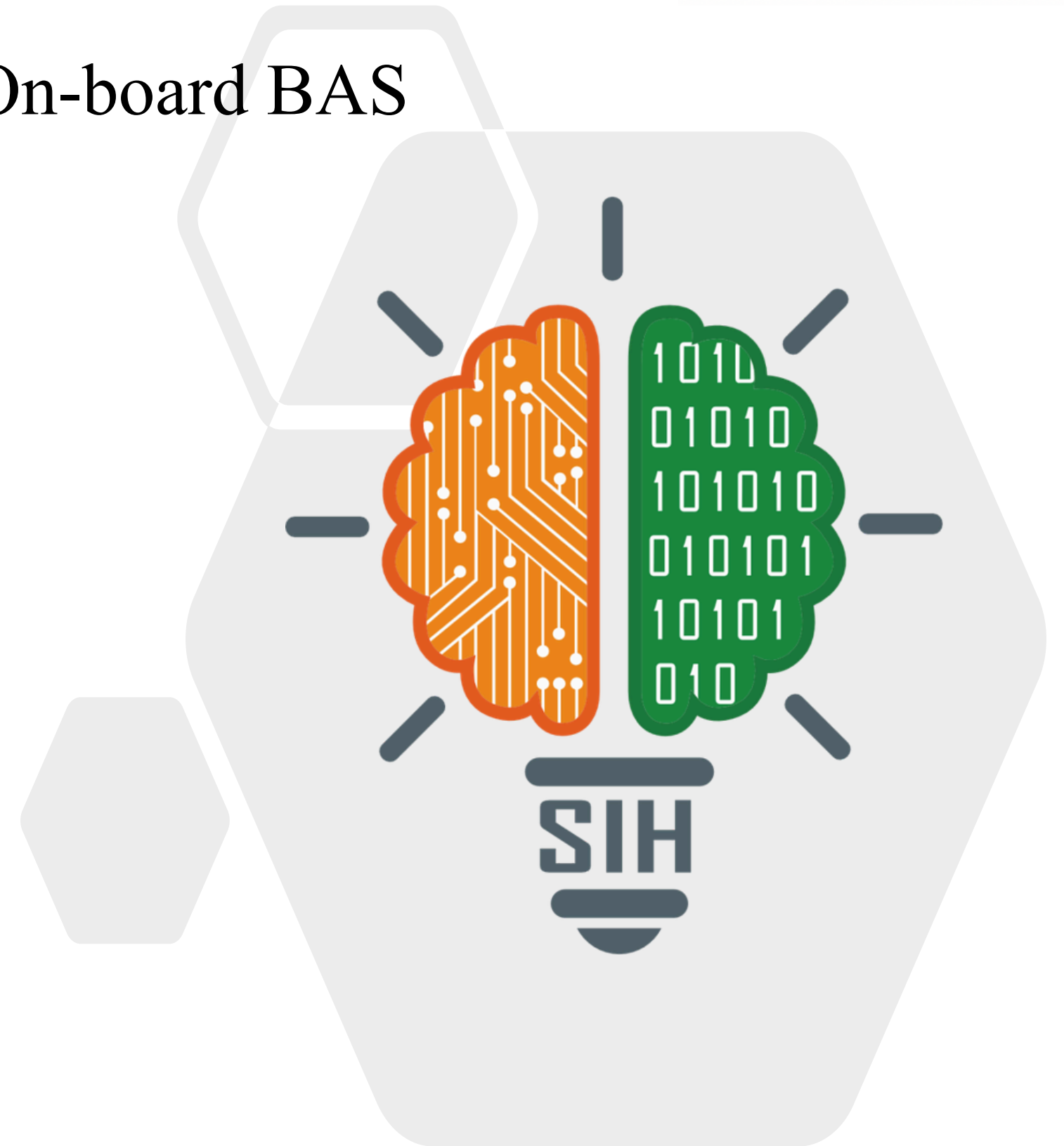


SMART INDIA HACKATHON 2026



AstroHAR: AI Human Activity Recognition for On-board BAS
Experiments

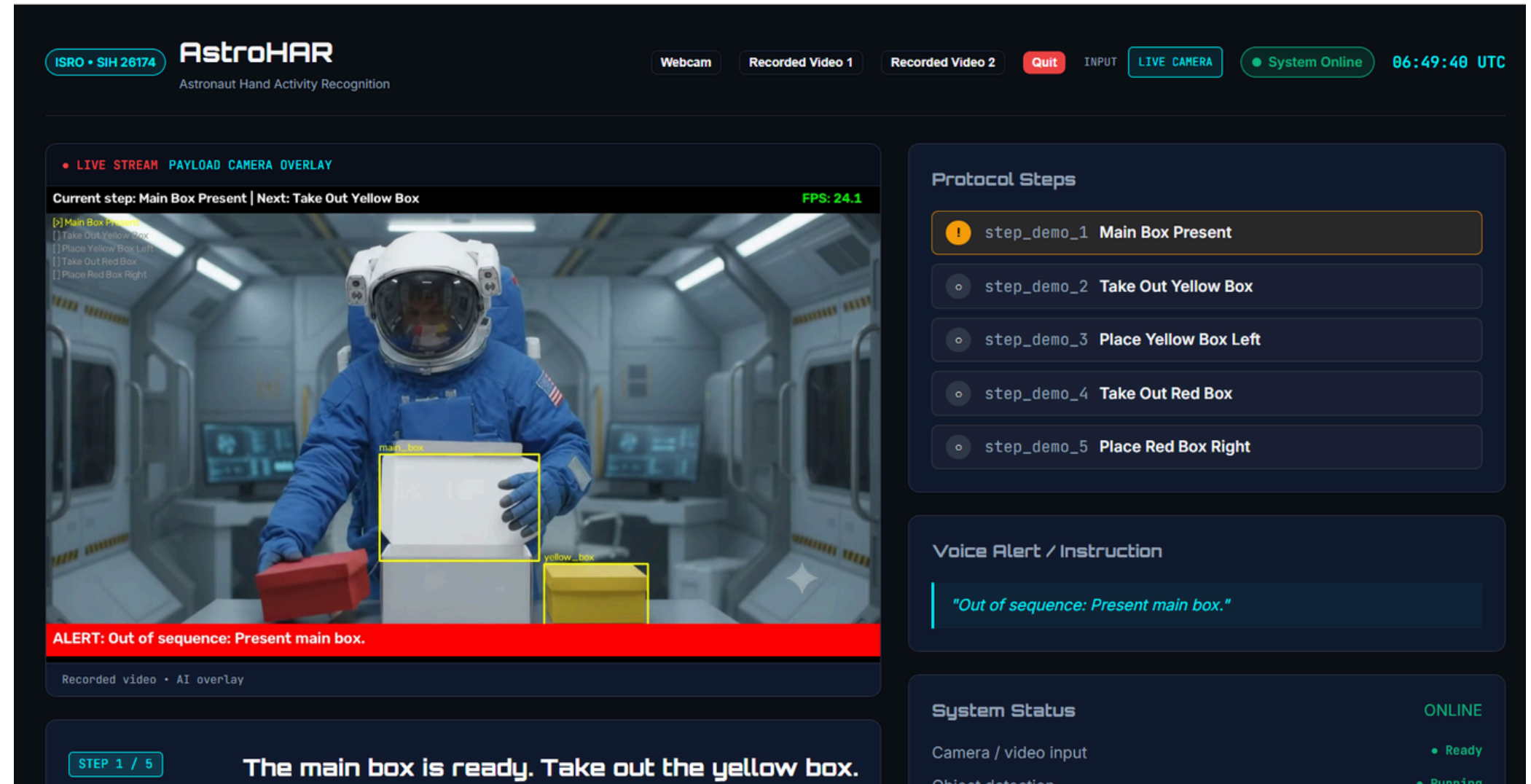
Problem Statement ID – 26174
Problem Statement – AI Human Activity
Recognition for On-board BAS
Experiments
Organization – ISRO
Theme – Space Technology
PS Category – Software



AI HUMAN ACTIVITY RECOGNITION FOR ON-BOARD BAS EXPERIMENTS

PROPOSED SOLUTION

- Offline local CPU video processing.
- Webcam + recorded-video input with source switching.
- OpenCV HSV + contour object detection.
- MediaPipe hand landmarks; hand-object contact & movement tracking.
- Five-state FSM validates:
 1. Main box present
 2. Take out yellow box
 3. Place yellow box on the left
 4. Take out red box
 5. Place red box on the right
- Voice instructions/alerts, wrong-placement and out-of-order detection.
- Timestamped JSONL logs + local MP4 recording.
- Browser-based local dashboard.



TECHNICAL ARCHITECTURE

Camera / Webcam / Recorded Video

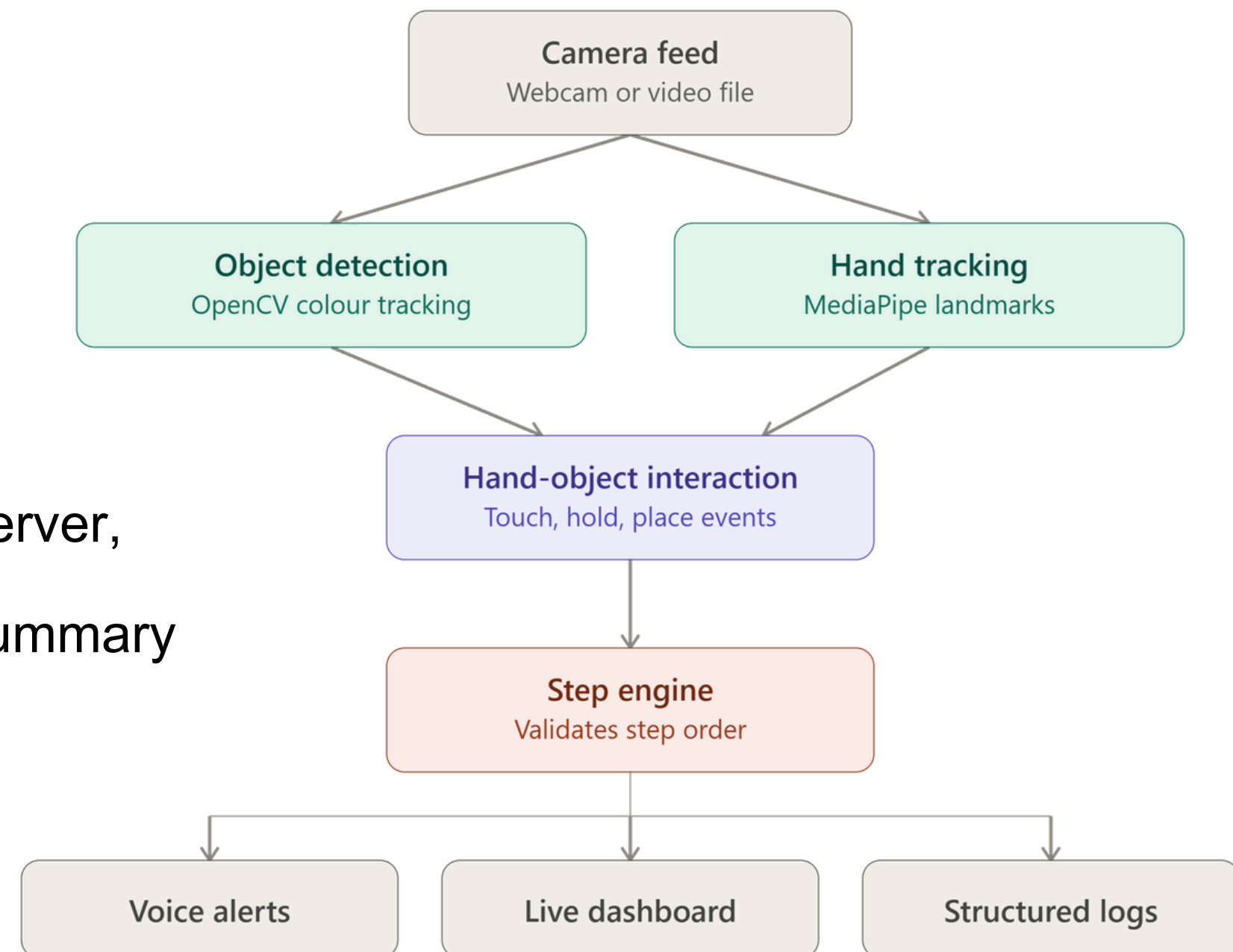
- Preprocessing
- OpenCV Object Detection
- MediaPipe Hand Landmarks
- Hand-Object Interaction
- Five-State FSM
- Voice Alerts + Dashboard + Logs + Local Recording

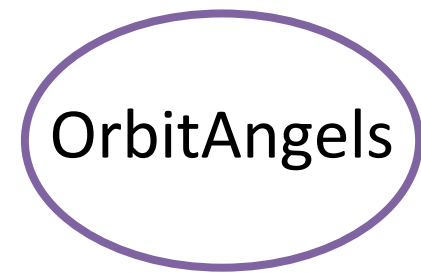
IMPLEMENTED

- CPU-based offline execution
- Browser-based local dashboard using Python HTTP server, HTML, CSS, and JavaScript
- Local MP4 + timestamped JSONL logs + end-of-run summary

PLANNED FUTURE UPGRADE

- YOLOv8/YOLO11
- RTMPose / full-body pose estimation
- GRU/TCN
- 3D Human Mesh Recovery





FEASIBILITY AND VIABILITY



CURRENT PROTOTYPE

- Local CPU execution; webcam + recorded video
- Local MediaPipe model; no internet during runtime
- Dashboard monitoring + voice guidance

ROADMAP

PHASE 1 — WORKING PROTOTYPE

OpenCV • MediaPipe hands • FSM • Dashboard • Voice alerts • Logging + recording

PHASE 2 — AI MODEL DEVELOPMENT

Dataset creation • Annotation • Lightweight object detection • Pose + interaction modeling

PHASE 3 — ADVANCED RESEARCH

GRU/TCN • ONNX optimization • Orientation-agnostic 3D HMR

FUTURE WORK

Labeled dataset

- Lightweight detector
- Full-body pose
- Confidence scoring
- Temporal action recognition
- ONNX/offline deployment

IMPACT AND BENEFITS



- Prevents skipped steps in the experiment sequence.
- Detects wrong placements and out-of-order actions.
- Provides voice-based correction and guidance.
- Reduces dependence on continuous ground control.
- Enables offline edge processing.
- Creates a timestamped experiment audit trail.
- Future use across BAS, lunar and other space experiments.

IMPLEMENTED REFERENCES

- OpenCV
- MediaPipe
- Python HTTP server
- pytsx3
- JSONL + MP4 local recording

FUTURE / ADVANCED FUTURE WORK

- YOLO / ONNX
- RTMPose
- GRU / TCN
- 3D Human Mesh Recovery

